

The main thing is not to oversalt — Solution

W21 (2021) — English translation

A1

?? pts

Measure the resistance of the limiting resistor R_1 with an ohmmeter (see equipment). The approximate value of 10 ohms is indicated below, but use the more accurate value you obtained in your calculations.

Let's measure the resistance of the limiting resistor $R_1 = 10,1 \, \Omega$ with a multimeter in ohmmeter mode.

A2

7.50 pts

Experimentally confirm the validity of the proposed model. To do this, study the current-voltage characteristics of the solution at 5 different distances between the electrodes.

We assemble a measuring circuit consisting of a battery connected in series, a control unit and a syringe with a salt solution. We connect a voltmeter to the electrodes of the syringe. We connect the second voltmeter in parallel with the reference resistance R_1 to measure the current in the circuit.

The entire set of processes on the cathode and anode is proposed to be represented in the form of a series circuit consisting of an external voltage applied to it U_0 , resistance of the electrolyte volume R , contact EMF U and contact resistance r (Fig. 1). It is assumed that the value R is directly proportional to the resistivity of the electrolyte ρ , the distance between the electrodes L and inversely proportional to the cross-sectional area of the electrolytic bath S .

The theoretical current-voltage characteristic of the electrolytic bath within the framework of this model has the form ρ and r_k .

We take the current-voltage characteristic of the solution. The tables show the results of measuring the current-voltage characteristics of the solution at 5 different volumes of liquid in the syringe V (and the corresponding distances L between the electrodes).

U_{10} , mV U , mV I , mA V , ml L , cm 290 1740 28.7 4 1.3 320 1860 31.7 450 2060 44.6 650 2460 64.4 770 2670 76.2 990 3080 98.0 1080 3420 106.9

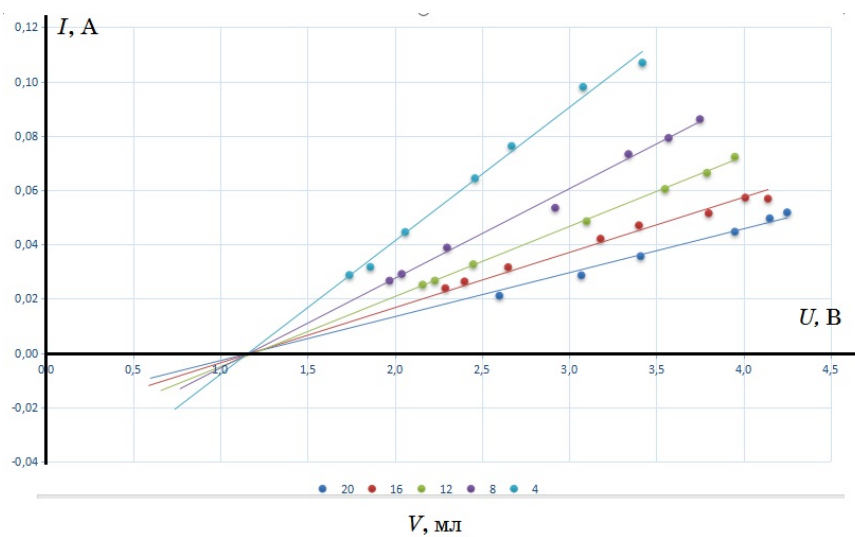
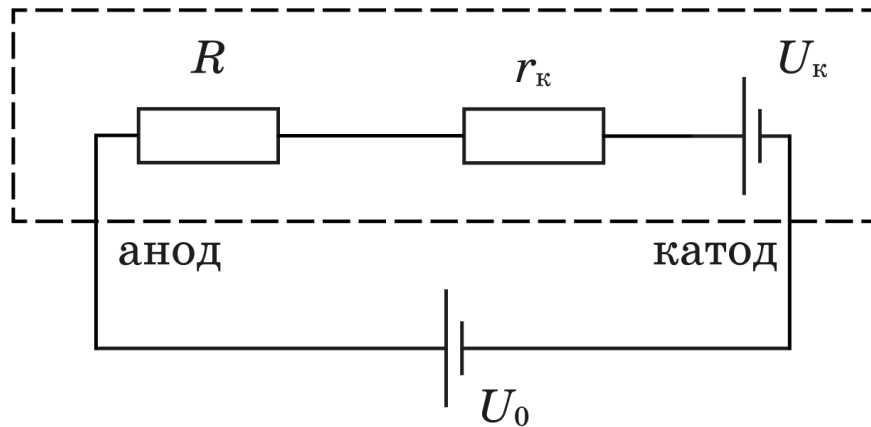
U_{10} , mV U , mV I , mA V , ml L , cm 870 3750 86.1 8 2.6 800 3570 79.2 740 3340 73.3 540 2920 53.5 392 2300 38.8 294 2040 29.1 269 1970 26.6

U_{10} , mV U , mV I , mA V , ml L , cm 254 2160 25.1 12 3.9 269 2230 26.6 330 2450 32.7 490 3100 48.5 610 3550 60.4 670 3790 66.3 730 3950 72.3

U_{10} , mV U , mV I , mA V , ml L , cm 574 4140 56.8 16 5.2 578 4010 57.2 520 3800 51.5 475 3400 47.0 425 3180 42.1 319 2650 31.6 266 2400 26.3

U_{10} , mV U , mV I , mA V , ml L , cm 213 2600 21.1 20 6.5 289 3070 28.6 360 3410 35.6 451 3950 44.7 501 4150 49.6 523 4250 51.8

We build the current-voltage characteristics for all experiments



A3

1.00 pts

Estimate the magnitude of the contact potential difference U_{to} .

Using the graph, we determine: 1. From the point of intersection of the graphs and the voltage axis, the value of the contact potential difference $U = 1,2$ V. 2. From the angular coefficient (see formula (2)) the resistance of the electrolyte volume together with the resistance of the contact areas R_{tot} for 5 values L .

A4

2.50 pts

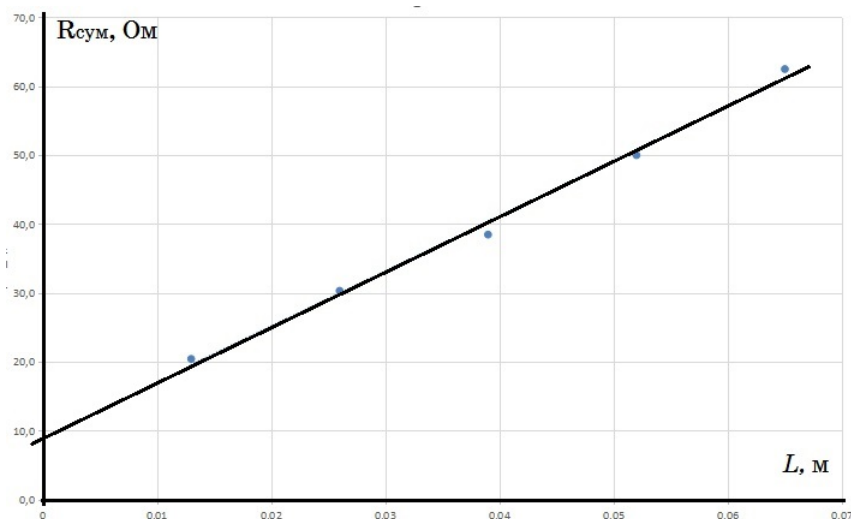
Determine the value of contact resistance r_{to} .

The results of processing experimental data for five values of L are presented in the table and figure.

No. L , m $1/R_{tot}$, $1/\text{Ohm}$ R_{tot} , Ohm

1	0.065	0.016	62.5
2	0.052	0.020	50.0
3	0.039	0.026	
4	0.026	0.033	30.3
5	0.013	0.049	20.4

From the graph we find $r = R_{tot}(0) = 9,2 \Omega$.



A5

1.00 pts

Calculate the specific conductivity σ of a three percent solution of sodium chloride in water.

Cross-sectional area of the piston $S = 3,1 \cdot 10^{-4} \text{ m}^2$ (can be found by measuring the diameter of the piston or the ratio of volume to length). Within the framework of the used model $R_{tot}/L = \rho/S$. From the slope of the straight line $R_{tot}(L)$ we find the resistivity of the salt solution used $\rho = 0,24 \text{ }\Omega \cdot \text{m}$. That is, $\sigma = 1/\rho = 4,2 (\Omega \cdot \text{m})^{-1}$.

A6

3.00 pts

Calculate the mobility μ of sodium and chlorine ions in this solution.

Specific conductivity of the electrolyte $\sigma = 1/\rho = q_1 n_1 + q_2 n_2$ where q_1 and q_2 are the modules of the electrical charges of positive and negative ions, respectively (in the case of sodium and chlorine ions $q_1 = q_2 = e$, where e is the elementary charge), n_1 and n_2 are their concentrations (in our case they are the same and equal to n), μ_1 and μ_2 are the mobilities of negative and positive ions in solution, which, according to the conditions of the problem, are also equal to each other $\mu_1 = \mu_2 = \mu$. Finally,

$$\sigma = \frac{2en}{\rho}$$

We calculate the concentration of sodium (or chlorine) ions in a three percent solution of table salt NaCl

$$n = N/V = \frac{m_{\text{NaCl}} N_A \rho}{M_{\text{NaCl}} m} = 0,03 \frac{N_A \rho}{M_{\text{NaCl}}} = 3,2 \cdot 10^{26} \text{ m}^{-3}$$

We calculate the mobility:

$$\mu = \frac{\sigma}{2en} = 4,0 \cdot 10^{-8} \frac{\text{m}^2}{\text{V} \cdot \text{s}}$$

, which is in good agreement with the literature data.